

HOMEWORK 8

1. Let p be a prime integer, $\xi \in \mathbb{C}$ primitive p -th root of unity ($\xi^p = 1, \xi \neq 1$). Find the degree of the extension $\mathbb{Q}(\xi + \xi^{-1})/\mathbb{Q}$.
2. Prove that $\mathbb{Q}(\sqrt{2} + \sqrt{3})/\mathbb{Q}$ is a normal extension.
3. Let p be a prime integer, $n, m > 0$. Find the smallest k such that the finite field $\mathbb{F}_{p^k}$ contains two subfields isomorphic to $\mathbb{F}_{p^n}$ and $\mathbb{F}_{p^m}$.
4. Find the degree of a splitting field of the polynomial $X^7 - 1$ over the finite field $\mathbb{F}_5$.
5. Let F be a field of characteristic $p > 0$ and E/F an extension of degree relatively prime to p . Show that E/F is a separable extension.
6. a) Let E/F be a **finite** field extension. Show that the set of all elements in E that are separable over F , is a subfield in E containing F . It is called the *maximal separable subextension in E/F* .
b) Let F be a field of characteristic 2. Find the maximal separable subextension in $F(X)/F(X^4 + X^2)$.
7. A finite field extension E/F is called *purely inseparable* if F is the maximal separable subextension in E/F . Prove that a finite field extension E/F is purely inseparable if and only if every field homomorphism $\sigma : F \rightarrow L$ has at most one extension $E \rightarrow L$.
8. Construct a Galois field extension $E/\mathbb{Q}$ of degree 15.
9. Determine the Galois group of the polynomial $X^6 + 3$ over $\mathbb{Q}$.
10. Let $K = \mathbb{Q}(\sqrt{3}, \sqrt[7]{5})$. Prove that K has only one subfield $F \subset K$ such that $[F : \mathbb{Q}] = 2$.